



# 8709 - Confronting Lyman-alpha Radiation Pressure Feedback in the JWST Era

Cycle: 4, Proposal Category: AR

## INVESTIGATORS

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## OBSERVATIONS

### ABSTRACT

Recent JWST observations have unveiled an unexpected abundance of UV-bright galaxies and dense star clusters at redshifts  $z > 10$ , challenging current models of galaxy formation during Cosmic Dawn. However, Lyman-alpha (Lya) radiation pressure is a strong feedback mechanism in these metal-poor environments, capable of regulating early star formation and black hole growth. Despite its potential importance, Lya feedback is not widely included in galaxy formation simulations due to computational challenges associated with on-the-fly Lya radiative transfer. We propose to confront Lya radiation pressure feedback by developing Lya radiation-hydrodynamic simulations and novel analytical solutions. Our objectives are to (i) develop numerical models to understand time-dependent outcomes, and (ii) design improved sub-grid models for cosmological simulations. We will manage computational challenges by pursuing both Monte Carlo Radiative Transfer and diffusion approximation solvers. Our systematic exploration across parameter spaces focuses on suppression mechanisms, critical column densities, velocity gradients, and delivery timescales. These efforts are crucial for interpreting JWST observations, given the high detection rate of Lya-emitting galaxies at  $z > 7$ , including spectroscopic confirmations at  $z > 13$ . By accurately accounting for Lya feedback, we will improve galaxy formation models with informed sub-grid recipes and feedback prescriptions, constrain star formation efficiencies, and enhance our understanding of the early universe during Cosmic Dawn.

### OBSERVING DESCRIPTION